
Specification-Conditioned Physics-Informed Neural Networks: Generalizing a Single Surrogate Across Coefficients, Initial, and Boundary Conditions



Abstract

Parameterized physics-informed neural networks amortize PINN training across a family of PDEs by encoding the equation coefficients into a latent vector that conditions a shared solution network. The coefficients, however, are only one axis of a problem specification: the initial condition (IC) and the boundary condition (BC) are imposed through the training loss and frozen into the trained weights, so the surrogate cannot be *queried* at a new IC or BC. We extend the coefficient encoder of such models into a *problem-specification encoder* that conditions jointly on the coefficients, the initial condition, and a typed boundary-condition embedding, holding the solution network, training objective, and budget of the underlying parameterized PINN fixed. On convection–diffusion–reaction benchmarks, one model trained once answers queries at unseen IC parameters, at IC families never seen in training, at unseen coefficient–IC combinations, and under three boundary-condition types (with an authored, convergence-validated non-periodic reference solver), at $1.6\text{--}4.6\times$ lower error than the frozen coefficient-only model—the reachable alternative without retraining—on every axis; corrupting the conditioning at query time collapses the gains, and latent diagnostics show genuine, smoothly organized conditioning. A capacity-matched control that ingests the same IC information as raw samples performs equivalently in distribution: contrary to the expectation we registered before the comparisons, the *representation* of the conditioning information is second-order at matched capacity—band-limited spectra and samples are linearly equivalent inputs—so *what* is conditioned on, not *how*, carries the generalization. Held-out error is rank-predicted (Spearman ≈ 0.95) by the IC’s spectral distance to the training set, and queries with out-of-band spectral content fail. That failure is a genuine boundary: it persists under a $2\times$ increase of encoder capacity, epochs, and collocation (which halves in-distribution error), and—in three registered attempts—could not be repaired by retraining on band-covering distributions, because with the solution network held fixed, coverage trades fit for span: the model cannot accurately fit rough specifications even when trained on them directly. The binding constraint is the shared solution pathway, not the conditioning interface. A single shared model further supports three operator families jointly, with never-trained operator compositions transferring at intermediate accuracy. We conclude that the problem specification—not just the coefficient vector—is the right conditioning variable for physics-informed surrogates, and that what the conditioning *covers* and what the shared solution network can *fit*, rather than the architectural form of the conditioning pathway, govern how far specification conditioning generalizes.

1 Introduction

Many applications of scientific computing require solving not one PDE but a family of them. Design optimization sweeps material coefficients; uncertainty quantification samples source profiles; inverse problems iterate over initial states; control problems vary boundary actuation. Classical solvers price each family member at the full cost of a new solve, and standard physics-informed neural networks (PINNs) [Raissi et al., 2019, Karniadakis et al., 2021] inherit this economics: one network is trained per problem, and training—not inference—is the dominant cost. This motivates *amortized* physics-informed surrogates that train once and answer many queries.

Parameterized PINNs (P²INNs) [Cho et al., 2024] are a strong recent instance of this program. They factor the surrogate $u_{\Theta}(x, t; \mu)$ into a coordinate encoder, a *parameter encoder* that maps the PDE coefficient vector μ to a latent code $\mathbf{h}_{\text{param}}$, and a shared decoder that reads the concatenated codes. Trained on a residual-based objective over a grid of coefficients, one model covers an entire coefficient range, mitigates well-documented PINN failure modes at large convection or reaction coefficients [Krishnapriyan et al., 2021], and interpolates to unseen coefficients without retraining.

But the coefficients are only one coordinate of the specification that defines a PDE problem. The initial condition (IC) and boundary condition (BC) enter such models exclusively through two loss terms—a data-matching term at $t = 0$ and a periodic boundary penalty—and are therefore *baked into* the trained weights. Nothing about the IC or BC is an input to the encoder. A P²INN trained with one initial profile cannot be evaluated at another; a model trained under periodic boundaries knows nothing of Dirichlet data. In exactly the settings that motivate amortization (varying sources, varying starting states, varying boundary types), the surrogate must be retrained per specification—the cost it was designed to avoid.

This paper asks whether the parameter encoder can be extended into a *problem-specification encoder*: a single conditioning pathway ingesting the coefficients, the initial condition, and the boundary condition, such that one trained model generalizes over all three axes—and what representation of the IC/BC makes such generalization work, rather than merely adding inputs. Three difficulties make this a research question rather than an engineering exercise. First, an initial condition is a *function*, not a vector; the encoder needs a finite representation that supports generalization to profiles it never saw, including profiles from different functional families (a Gaussian pulse is not a sinusoid). Second, naively appending sampled IC values to the encoder input adds capacity and information at once, so any gain must be disentangled from the extra inputs themselves. Third, for non-periodic boundary conditions the training signal itself changes (the boundary term of the loss must express the conditioned BC), and trustworthy reference solutions must exist to measure generalization at all.

We address these with a deliberately minimal design that isolates the conditioning question. Keeping the P²INN decoder, coordinate encoder, injection interface, four-term loss, optimizer, and budget fixed, we replace only the parameter encoder with a factorized specification encoder: a coefficient branch; an IC branch that reads the leading Fourier modes of the initial profile (computed from the same 64 samples a raw-input encoder would see); an optional BC branch reading a boundary-type embedding and boundary values; and a fusion head producing the same 50-dimensional $\mathbf{h}_{\text{param}}$. As controls we train (i) the original coefficient-only model and (ii) a *raw-input* variant that receives the identical 64 IC samples (and BC vector) as a flat input through an encoder whose parameter count matches ours to within 1%. Any advantage of our encoder over the raw variant is then attributable to the *structure* of the conditioning, not its capacity or information content.

Contributions.

- **A specification encoder for parameterized PINNs.** We extend the coefficient encoder of P²INN to condition jointly on the PDE coefficients, the initial condition, and a typed boundary embedding—with everything else held fixed—and show that a single model then answers queries across the specification space: unseen IC parameters, unseen IC functional families, unseen coefficient $\times$ IC and IC $\times$ BC combinations, and three BC types, at a fraction of the frozen coefficient-only model’s error, with seed-robust margins and near-preserved seen-specification fidelity.
- **Attribution at matched budget—with a negative result we did not expect.** Query-time corruption of the conditioning collapses the gains and the latent code demonstrably encodes the IC, so the conditioning is causal; but the spectral, factorized encoder does *not* beat the

capacity-matched raw-input control in distribution, on any of three families, under margins we fixed in advance. We trace this to a linear equivalence of band-limited representations and treat it as a finding: at matched capacity and information, conditioning *content* governs generalization, not its in-band parameterization.

- **A mechanism, verified where it bites.** The encoder embeds initial conditions by their low-order spectrum (phase circles, amplitude rays, out-of-family profiles interpolated in-manifold), and a single specification-space statistic—spectral distance to the training set—rank-predicts held-out error (Spearman ≈ 0.95). We then push the mechanism to a genuine generalization boundary: failures at out-of-band queries persist under a $2\times$ encoder-capacity/epoch/collocation increase that halves in-distribution error, and three registered band-coverage retraining attempts fail to repair them—at fixed solution network, coverage trades fit for span, and the model cannot accurately fit rough specifications even trained on them directly. The boundary is a solution-capacity limit sharpened by conditioning coverage, and the diagnosis is itself mechanism-consistent (the failed first coverage distribution covered the band only nominally, exactly as the spectral statistic predicts).
- **Boundary conditions and operator breadth.** With an authored, convergence-validated Crank–Nicolson reference, one model conditions on periodic, Dirichlet, and zero-flux Neumann boundaries, generalizing to withheld (coefficient, IC, BC) pairings at seen-level accuracy; a single shared model also supports convection, reaction, and reaction–diffusion operators jointly and transfers at intermediate accuracy to operator compositions (convection–diffusion, full CDR) never trained in any form.

2 Related work

Amortized and parameterized PINNs. Beyond per-problem PINNs [Raissi et al., 2019], several lines reduce the cost of solving PDE families: curriculum and failure-mode analyses at hard coefficients [Krishnapriyan et al., 2021], meta-learned initializations [Liu et al., 2022, Penwarden et al., 2023], hypernetwork-generated low-rank PINNs [Cho et al., 2023b], and separable architectures [Cho et al., 2023a]. P²INN [Cho et al., 2024] conditions a shared PINN on the coefficient vector through a dedicated encoder and is the platform for this study; the Meta-Auto-Decoder [Huang et al., 2022] learns a task latent per problem by optimization at test time, which in principle spans IC/BC variability but requires per-task adaptation rather than zero-shot conditioning. Our question is orthogonal to all of these: not *how* to train the shared model, but *what* it should be conditioned on, and which representation of the non-coefficient specification generalizes.

Operator learning. DeepONet [Lu et al., 2021, Chen and Chen, 1995] and Fourier neural operators [Li et al., 2021, Kovachki et al., 2023] learn maps from input functions (often the IC) to solutions, typically from simulation data; physics-informed variants train the operator on the PDE residual [Wang et al., 2021, Li et al., 2024]. Our raw-input control is deliberately DeepONet-flavored—an encoder fed sampled function values—so the comparison quantifies what a spectral, factorized specification representation adds over the standard raw-sample branch under identical physics-informed training and capacity. Distinct from the operator setting, our model conditions *jointly* on coefficients, IC, and BC type (operators are usually trained per coefficient setting and per BC), and our analysis targets the representational question of *which* encodings of the specification generalize, including across functional families, boundary types, and operator compositions. FNO’s spectral convolutions and Fourier-feature coordinate embeddings [Tancik et al., 2020] both motivate our choice of spectral IC features, but here the spectrum is used as a *conditioning representation* for a pointwise PINN surrogate, not as a layer parameterization or coordinate embedding; and input/output representation variants are known to matter empirically in data-driven operator learning [Lu et al., 2022], whereas our matched-capacity result isolates when they provably cannot (linearly equivalent band-limited inputs) under physics-informed training.

Conditional neural fields and multi-physics surrogates. Viewing the solution as a neural field modulated by a context [Dupont et al., 2022, Hagnberger et al., 2024, Wang et al., 2024], conditioning on PDE parameters has been explored with attention and modulation mechanisms [Takamoto et al., 2023], and data-driven multi-operator pretraining spans several physical systems at once [McCabe et al., 2024, Subramanian et al., 2023, Ye et al., 2024]. These are data-rich, solver-supervised settings; in contrast, our model is trained purely from the residual and the conditioned IC/BC terms, at the scale of a single P²INN, and our contribution is the controlled analysis of specification conditioning—

capacity-matched raw-input control, conditioning ablations, and a spectral-coverage account of where generalization stops—rather than scale or a new backbone.

3 Setting: from coefficient conditioning to specification conditioning

The benchmark family. We study the 1D convection–diffusion–reaction (CDR) family on the periodic domain $x \in [0, 2\pi)$, $t \in [0, 1]$,

$$\frac{\partial u}{\partial t} + \beta \frac{\partial u}{\partial x} - \nu \frac{\partial^2 u}{\partial x^2} - \rho u(1 - u) = 0, \quad u(x, 0) = u_0(x), \quad (1)$$

with Fisher reaction term $\rho u(1 - u)$, following the benchmark construction of Krishnapriyan et al. [2021] and Cho et al. [2024]. Setting coefficients to zero specializes (1) to convection (β), diffusion (ν), reaction (ρ), and their combinations. A *problem specification* is the tuple

$$s = (\boldsymbol{\mu}, u_0, b), \quad \boldsymbol{\mu} = (\beta, \nu, \rho), \quad (2)$$

where u_0 is the initial condition and b the boundary condition. The reaction subfamily is central to our evaluation design: its solution map $u_0 \mapsto u(\cdot, t)$ is *nonlinear* (the exact solution is the pointwise logistic flow $u = u_0 e^{\rho t} / (1 - u_0 + u_0 e^{\rho t})$), so conditioning on the IC requires representing how a whole profile deforms, whereas for pure convection/diffusion the solution is a single linear operator applied to u_0 and IC generalization is comparatively easy. We therefore always demonstrate IC generalization on a family with nonlinear IC dependence (reaction, and reaction–diffusion for the boundary study), and use convection as breadth.

The platform. P²INN [Cho et al., 2024] factorizes the surrogate as

$$u_\Theta(x, t; \boldsymbol{\mu}) = g_{\theta_g}([g_{\theta_c}(x, t); g_{\theta_p}(\boldsymbol{\mu})]), \quad (3)$$

with a coordinate encoder $g_{\theta_c}: (x, t) \mapsto \mathbf{h}_{\text{coord}} \in \mathbb{R}^{50}$, a parameter encoder $g_{\theta_p}: \boldsymbol{\mu} \mapsto \mathbf{h}_{\text{param}} \in \mathbb{R}^{50}$ (an MLP 3–150–150–150–50 with tanh), and a residual decoder g_{θ_g} over the concatenation, which additionally receives a linear “hyper-shift” of $(\boldsymbol{\mu}, x, t)$ at its last hidden layer. All three are trained jointly on

$$\mathcal{L} = \underbrace{\text{MSE}(\mathcal{R}[u_\Theta])_\Omega}_{\text{residual, collocation}} + \underbrace{\text{MSE}(\mathcal{R}[u_\Theta])_{t=0}}_{\text{residual, IC points}} + \underbrace{\text{MSE}(u_\Theta - u_0)_{t=0}}_{\text{IC data term}} + \underbrace{\mathcal{L}_b}_{\text{boundary term}}, \quad (4)$$

with equal weights, where $\mathcal{R}$ is the residual of (1) and $\mathcal{L}_b$ penalizes $u_\Theta(0, t) \neq u_\Theta(2\pi, t)$ in the periodic case. The specification enters (4) in exactly two places—the IC data term and the boundary term—and enters the *model* (3) nowhere. This is the representational gap we close: for a fixed trained model, u_0 and b are constants of the loss, not queryable inputs.

4 A problem-specification encoder

Our design principle is to change *only* what the parameter encoder is conditioned on, so that any change in generalization behavior is attributable to the conditioning. The decoder, coordinate encoder, concatenation injection, hyper-shift interface (which continues to receive $(\boldsymbol{\mu}, x, t)$ only, for every model variant), loss terms, weights, collocation counts, optimizer, and epochs are those of Cho et al. [2024], unchanged.

Representing the initial condition. Every model variant observes the initial condition through the same measurement: its values $\mathbf{u}_0 = (u_0(x_1), \dots, u_0(x_{64}))$ at 64 equispaced points. The question is how the encoder should consume them. Our encoder applies a fixed, invertible-up-to-truncation linear map first: the real DFT of $\mathbf{u}_0$, truncated to the leading K modes and scaled to amplitude units,

$$\phi(u_0) = (c_0, \text{Re } c_1, \dots, \text{Re } c_K, \text{Im } c_1, \dots, \text{Im } c_K) \in \mathbb{R}^{2K+1}, \quad c_k = \frac{2^{[k>0]}}{64} \sum_j u_0(x_j) e^{-ikx_j}. \quad (5)$$

Three properties motivate this choice over raw samples. (i) *Dynamics alignment*: for the linear part of (1) the evolution is diagonal in exactly these coordinates (mode k evolves by $e^{(-\nu k^2 - i\beta k)t}$), and the reaction nonlinearity acts smoothly on low-order mode interactions, so the map from $\phi(u_0)$ to the solution is smooth and low-dimensional for smooth data. (ii) *Factorization of nuisance*: translations of

the profile move phase but not amplitude, amplitude scalings act linearly—families that are “different functions” pointwise (an offset sinusoid, a wide Gaussian, a parabolic bump) become nearby points in a shared, smoothly organized coordinate system. (iii) *Explicit truncation*: the representation declares which spectral band the encoder can see, making the model’s generalization envelope analyzable (§6.3, §6.4).

Representing the boundary condition. A BC is encoded as a type one-hot over {periodic, Dirichlet, Neumann zero-flux} plus the two boundary values (g_L, g_R) (zero unless Dirichlet). During training, the boundary term $\mathcal{L}_b$ of (4) is evaluated *for the conditioned BC of each specification*: the reference periodic pair penalty, $\text{MSE}(u_\Theta(0 \vee 2\pi, t) - g_{L \vee R})$ for Dirichlet, and $\text{MSE}(\partial_x u_\Theta(0 \vee 2\pi, t))$ for Neumann, pooled over boundary points.

The encoder. The specification encoder is factorized:

$$\mathbf{h}_{\text{param}} = f([m(\boldsymbol{\mu}); e(\phi(u_0)); q(b)]), \quad (6)$$

with branch MLPs $m : 3 \rightarrow 64 \rightarrow 64$, $e : (2K+1) \rightarrow 64 \rightarrow 64$, $q : 5 \rightarrow 16 \rightarrow 16$ (all tanh; q present only in the BC study) and a fusion head $f : \rightarrow 150 \rightarrow 150 \rightarrow 50$ matching the reference encoder’s output dimension, so $\mathbf{h}_{\text{param}}$ feeds the unchanged concatenation. All conditioning inputs travel a float path.

Controls. Two comparators isolate what matters:

- *Coefficient-only* (the reference P²INN): the verbatim 3–150–150–150–50 encoder, trained on its fixed-IC (and periodic) setting. It cannot be queried at a new IC/BC; when we evaluate it on held-out specifications we evaluate it *frozen*—its answer is its trained-specification solution—which is precisely the accuracy a practitioner obtains from it without retraining.
- *Raw-input conditioning*: the same information, no structure. Input $[\boldsymbol{\mu}; \mathbf{u}_0; b] \in \mathbb{R}^{67(+5)}$ feeds a flat MLP of the reference encoder’s shape whose width is solved so its encoder parameter count matches ours to < 1% (60,480 vs. 60,302 without the BC branch). It sees the same 64 samples the spectral features are computed from (strictly more information than the $2K+1$ truncation retains); it lacks only the representation and the factorization.

Both IC-conditioned variants are trained, per experiment, on identical specification sets, points, seeds, and budget.

5 Experimental setup

Protocol fidelity. We reuse the reference pipeline wholesale: its point construction (1000 collocation, 256 IC, 100 boundary points per specification, drawn with the reference sampler), its solvers (exact logistic reaction; exact spectral convection–diffusion; operator-split reaction–diffusion/CDR), its metric (per-specification relative L_2 error over 1000 fixed test points and mean absolute error, averaged over specifications), its optimizer (Adam, default learning rate), its budget (20,000 full-batch epochs), and 3 seeds per configuration with mean $\pm$ std reported. Our generalization of the solvers to arbitrary IC callables is validated bit-for-bit against the originals, and our full pipeline reproduces the released P²INN checkpoint’s published per-family errors to four decimal places (Appendix E). Reported wins must exceed a margin fixed before the comparisons: an axis counts as improved only if the mean relative L_2 error drops by at least $\max(0.015, 15\% \text{ of the comparator})$ and the margin exceeds the pooled seed standard deviation.

Specification splits. All splits were fixed, with seeds and exact lists, before any comparison ran (they ship with our code). For the core study on *reaction* ($\rho \in \{1, \dots, 5\}$): 24 trained ICs from the offset-sinusoid family $u_0 = c + a \sin(\omega x + \varphi)$ ($c \in [0.35, 0.65]$, $a \in [0.10, 0.30]$, $\omega \in \{1, 2\}$, $\varphi \in [0, 2\pi)$; positivity-constrained for the Fisher term), each trained with only 3 of the 5 coefficients (the remaining 46 pairs form the *unseen-combination* set); 12 fresh sinusoids form the *unseen-IC-parameter* set; and 8 profiles from families never trained in any form—Gaussians of three widths and two amplitudes, and parabolic bumps—form the *unseen-family* set, all evaluated at every ρ . One anchor IC ($0.5 + 0.25 \sin x$) is trained at all five coefficients and doubles as the coefficient-only model’s fixed IC, so seen-specification fidelity is compared per-coefficient on common ground. The convection study mirrors this with $\beta \in \{1, \dots, 5\}$, larger-amplitude signed sinusoids, and anchor $1 + \sin x$ (a published reference setting). A third core study on *reaction–diffusion* ($\nu, \rho \in \{1, 2, 3\}^2$, periodic), structurally identical to the reaction study, was declared as a registered amendment after

Table 1: **Main results: relative L_2 error (mean $\pm$ std over 3 seeds) across the specification space.** “Seen” is each model’s trained-specification accuracy (for the coefficient-only model: its fixed-IC anchor setting, the original P²INN configuration). On the held-out axes the coefficient-only model cannot represent the queried IC and is evaluated frozen (its answer is its trained-IC solution)—the accuracy actually reachable without retraining. Unseen IC parameters: fresh draws from the trained sinusoid family; unseen IC family: Gaussian and parabolic-bump profiles never trained in any form; unseen combinations: coefficient–IC pairs whose members are each trained but never jointly. Both conditioned models share capacity to $< 1\%$ and the identical training recipe. Absolute-error version in Table 4.

Method	Family	Seen spec.	Unseen IC params	Unseen IC family	Unseen combos	Budget
P ² INN (coefficient-only, frozen)	reaction	0.0034 $\pm$ 0.0020	0.2488 $\pm$ 0.0008	0.6085 $\pm$ 0.0026	0.2245 $\pm$ 0.0008	reference
raw-input conditioning (control)	reaction	0.0055 $\pm$ 0.0008	0.0909 $\pm$ 0.0066	0.3444 $\pm$ 0.0476	0.0247 $\pm$ 0.0091	matched
spectral specification encoder (ours)	reaction	0.0099 $\pm$ 0.0039	0.1088 $\pm$ 0.0075	0.3792 $\pm$ 0.0137	0.0553 $\pm$ 0.0129	matched
P ² INN (coefficient-only, frozen)	reaction–diffusion	0.0680 $\pm$ 0.0052	0.2064 $\pm$ 0.0036	0.5297 $\pm$ 0.0055	0.2189 $\pm$ 0.0038	reference
raw-input conditioning (control)	reaction–diffusion	0.1044 $\pm$ 0.0023	0.0897 $\pm$ 0.0056	0.2290 $\pm$ 0.0401	0.1007 $\pm$ 0.0021	matched
spectral specification encoder (ours)	reaction–diffusion	0.1003 $\pm$ 0.0023	0.0842 $\pm$ 0.0058	0.2656 $\pm$ 0.0147	0.0994 $\pm$ 0.0016	matched
P ² INN (coefficient-only, frozen)	convection	0.0075 $\pm$ 0.0028	1.5125 $\pm$ 0.0033	2.5121 $\pm$ 0.0067	1.5006 $\pm$ 0.0032	reference
raw-input conditioning (control)	convection	0.0361 $\pm$ 0.0044	0.6249 $\pm$ 0.0119	1.1123 $\pm$ 0.0235	0.3792 $\pm$ 0.0493	matched
spectral specification encoder (ours)	convection	0.0388 $\pm$ 0.0087	0.6215 $\pm$ 0.1067	1.1036 $\pm$ 0.0988	0.4553 $\pm$ 0.0499	matched

the reaction/convection outcomes and before any of its runs (its rationale is part of the findings, §6.2); its operator-split references are refinement-checked at the sharpest evaluated profiles (changes $\leq 2.1 \times 10^{-3}$). A disjoint *stress probe* set (sinusoids at $\omega \in \{3, 4, 6\}$, narrow Gaussians $\sigma \in \{\pi/8, \pi/16\}$, and saturated sinusoids) supports the out-of-distribution analysis of §6.4 and is excluded from all headline claims.

Boundary-condition study. On reaction–diffusion ($\nu, \rho \in \{1, 2, 3\}^2$) we train one model across periodic, Dirichlet (boundary values = IC endpoint values), and zero-flux Neumann conditions, using cosine-type ICs $c \pm a \cos(\omega x)$ compatible with all three types. Each (coefficient, IC) pair is trained under two of the three BC types by a fixed rotation; the withheld third pairings (99 specifications) test BC-composition generalization, and 6 fresh ICs (with correspondingly novel Dirichlet boundary values) test IC generalization under every BC. Because the reference solvers are periodic, we author a Crank–Nicolson solver with the reference’s exact-logistic Lie splitting for the non-periodic cases and validate it two ways: agreement with the spectral reference on the periodic overlap ($\leq 3.1 \times 10^{-4}$ max-relative deviation) and self-convergence under space–time refinement (deviations $\leq 4.3 \times 10^{-3}$, decreasing first-order; Appendix F). Operator-split references are refinement-checked at the sharpest evaluated specifications (all changes $< 10^{-3}$).

Shared-model breadth study. A single model is trained jointly on convection ($\beta \leq 5$), reaction ($\rho \leq 5$), and reaction–diffusion ($\nu, \rho \leq 2$) specifications—the coefficient vector itself encodes the operator (zeros mark absent terms)—and evaluated within each family on unseen ICs and combinations, and on two operator compositions never trained in any form: convection–diffusion and full CDR.

Hyperparameters. The only tuned quantity is the spectral truncation $K \in \{4, 8, 16\}$, selected on validation points of *trained* specifications only (never on held-out sets), with a parsimony tie-break (smallest K within 10% of the best); $K=16$ was selected (Appendix G) and used everywhere, with $K=4$ retained as a disclosed band-limited diagnostic for §6.4. Branch widths were fixed a priori. The added encoder capacity over the reference (53,450 $\rightarrow$ 60,302 parameters, +13%) is granted equally to the raw-input control; decoder and coordinate encoder (23,401 parameters) are identical everywhere.

6 Results

6.1 A single conditioned model answers held-out specification queries

Table 1 and Figure 1 give the core result. On the reaction family, the specification-conditioned model reaches 0.0099 relative L_2 error across its 74 trained specifications—within $3\times$ of the coefficient-only specialist’s 0.0034 on its single trained IC—and, with no retraining, answers queries at unseen IC

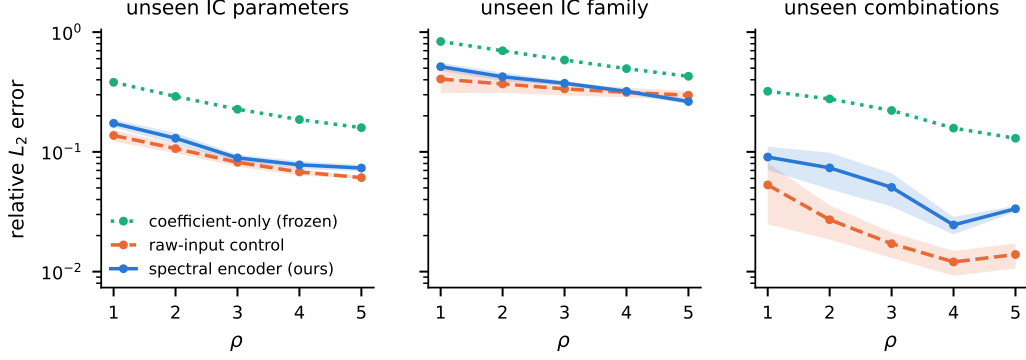


Figure 1: **Error across the specification space (reaction family).** Mean relative L_2 error per reaction coefficient on each held-out axis (bands: ± 1 std over 3 seeds; log scale). Both conditioned models track the reference across held-out initial conditions at a fraction of the frozen coefficient-only model’s error; errors decrease with ρ as the logistic flow contracts solutions toward $u=1$. The convection analogue is in Appendix C.

parameters at 0.1088, at initial-condition families never seen in training at 0.3792, and at unseen coefficient–IC combinations at 0.0553. The frozen coefficient-only model, the reachable alternative at equal training cost, incurs 0.2488, 0.6085, and 0.2245 on the same queries: the conditioned model reduces error by $2.3\times$ on unseen IC parameters, $1.6\times$ on unseen families, and $4.1\times$ on unseen combinations, with margins (0.1400, 0.2293, 0.1692) that exceed the pre-specified minimum improvement ($\max(0.015, 15\% \text{ of the comparator})$) several-fold and the pooled seed standard deviations (≤ 0.0139) by an order of magnitude. The same holds on reaction–diffusion (0.0842 vs. 0.2064 frozen on unseen ICs) and on convection, where transport makes conditioning much harder for everyone (0.6215 vs. 1.5125 frozen), and on every axis of the boundary-condition study below. Unseen *combinations* are the easiest axis (reaction: 0.0553, approaching seen-specification fidelity), unseen *families* the hardest—a gradient §6.3 explains quantitatively.

What conditioning costs at the anchor. The strictest reading of “no seen-specification regression”—per-coefficient error at the shared anchor IC within $\max(10\%, 0.005)$ of the specialist’s—holds for the reaction model at $\rho \geq 3$ but is exceeded by up to 0.0024 relative L_2 error at $\rho \in \{1, 2\}$ (spectral arm; the raw-input arm passes at all ρ), and is exceeded at 4 of 5 convection coefficients by both conditioned arms (worst case 0.0233 vs. specialist 0.0076) and at the harder reaction–diffusion anchors by both arms (0.0997 vs. specialist 0.0680). Covering 74–124 specifications at the specialist’s exact budget is not free (Appendix B tabulates every cell); anchor errors nonetheless stay within $3\times$ of the specialist’s—below 0.024 relative L_2 error on reaction and convection and 0.11 on reaction–diffusion—versus 0.21–2.5 for any cross-IC query the specialist itself cannot serve.

6.2 Attribution: the information is causal; in-band, its representation is second-order

Three tests establish that the gains come from genuine conditioning on the initial condition. (i) *Corrupting the conditioning removes the gains* (Figure 3): at query time, shuffling which IC representation accompanies which specification degrades unseen-IC-parameter error from 0.1088 to 0.2738, and replacing it with a constant (the mean trained representation) to 0.2073—at or beyond the frozen baseline’s 0.2488; on unseen families the ablations degrade 0.3792 to 0.4531 and 0.4130. (ii) *The encoding is not degenerate*: across trained ICs the IC-branch embedding retains 85% relative spread around its mean (0.852), a ridge probe recovers the leading spectral coordinates of u_0 from it ($R^2 = 0.803$ on trained, 0.502 on held-out ICs), and pairwise latent distances track pairwise spectral distances (Spearman 0.816). (iii) *The dependence is on the specific IC*, not on any added input: distinct ICs map to distinct codes that move the prediction exactly as the reference solution moves (Figure 2).

The representation surprise. We expected the spectral, factorized encoder to beat the raw-input control on held-out ICs; it does not. Under the pre-specified margin, the two are statistically tied on every in-band axis of every family—raw inputs are, if anything, slightly better on reaction (unseen

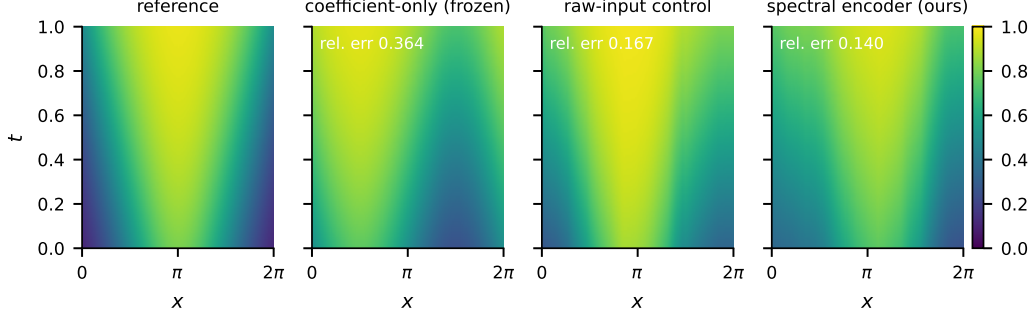


Figure 2: **A held-out-family query.** Reference and predictions for a Gaussian initial condition (never trained; $A=0.8$, $\sigma=\pi/2$) under reaction $\rho=2$. The frozen coefficient-only model can only reproduce its trained sinusoid’s solution; both conditioned models reconstruct the correct profile from the IC input alone, seed 0 shown.

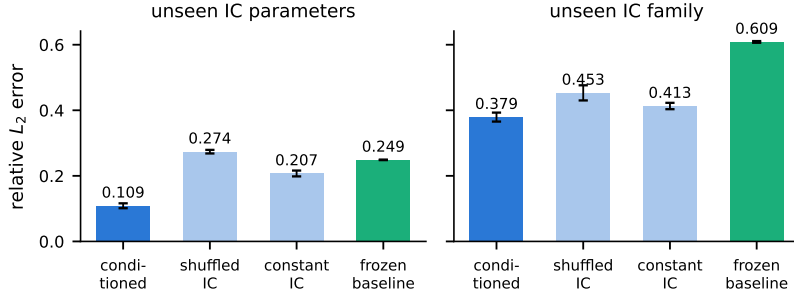


Figure 3: **Conditioning ablations (reaction, spectral encoder).** Replacing the IC input of the trained model at query time with another specification’s IC (“shuffled”) or the mean trained IC (“constant”) collapses held-out accuracy to—or beyond—the frozen coefficient-only level. Bars: mean over 3 seeds ($\pm$ std). The raw-input control behaves identically (Appendix C).

combinations 0.0247 vs. 0.0553; unseen parameters 0.0909 vs. 0.1088), and indistinguishable on convection and reaction–diffusion (Table 1). We registered the reaction–diffusion core study *after* observing this, predicting spatial coupling might separate the representations; it did not (0.0842 vs. 0.0897). In retrospect the tie has a clean explanation, which we regard as a finding rather than a failure. For band-limited initial conditions the truncated-spectral features are an invertible *linear* transform of the 64 raw samples—information the first linear layer of either encoder can absorb—so in-band the two arms present the same function class to the same optimizer; and for the reaction-family in particular, the solution map is *pointwise* in u_0 , so raw samples are already the dynamics-aligned coordinates. What the specification encoder is conditioned *on* (which information reaches $\mathbf{h}_{\text{param}}$) governs generalization; *how* that information is parameterized is second-order at matched capacity. Indeed even a deliberately band-limited encoder ($K=4$, unable to represent modes above 4) behaves indistinguishably on every axis we measure (unseen parameters 0.1247 vs. 0.1088)—in this regime nothing ever exploits the extra band, for reasons §6.4 makes precise. Practically, the tie licenses the simplest adequate conditioning interface; scientifically, it relocates the design question from the conditioning architecture to what the training distribution *covers* and what the shared solution network can *fit*.

6.3 Mechanism: the latent organizes by spectrum, and spectral distance predicts error

Figure 4 connects the representation to the generalization behavior. The left panel shows the encoder has learned the natural coordinates of the IC space: phase is a circle, amplitude a ray, and—crucially—Gaussians and bumps, pointwise unlike any trained profile, land *inside* the sinusoid manifold at positions consistent with their leading Fourier coefficients. Generalization to unseen

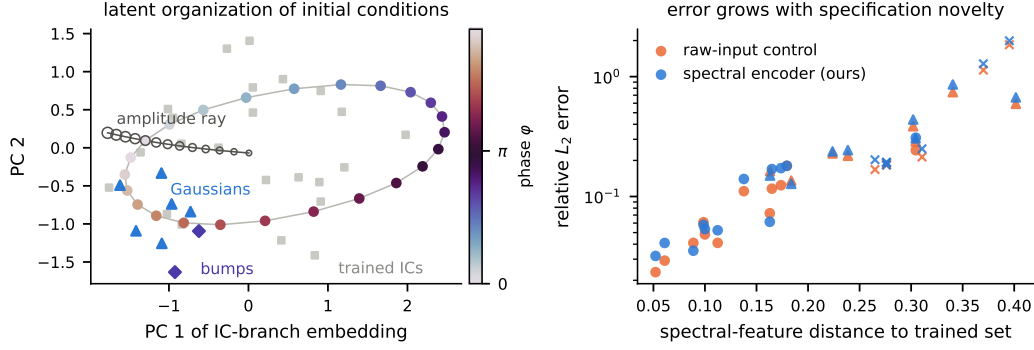


Figure 4: **What the encoder represents and what it predicts (reaction, seed 0).** Left: the IC-branch embedding, projected on the trained ICs' two leading principal directions. A phase sweep $0.5 + 0.25 \sin(x + \varphi)$ traces a closed loop ordered by φ ; an amplitude sweep traces a transverse ray (marker size $\propto$ amplitude); held-out Gaussians ($\blacktriangle$) and bumps ($\blacklozenge$) embed inside the trained region near spectrally similar sinusoids—the encoder coordinates ICs by their low-order spectrum, which is why profiles from unseen functional families are answerable at all. Right: mean relative L_2 error of every held-out and probe IC (circles: unseen parameters; triangles: unseen family; crosses: probes) against its spectral-feature distance to the trained set; both arms collapse onto one monotone curve (Spearman $\rho_s=0.958$ ours, 0.953 raw control).

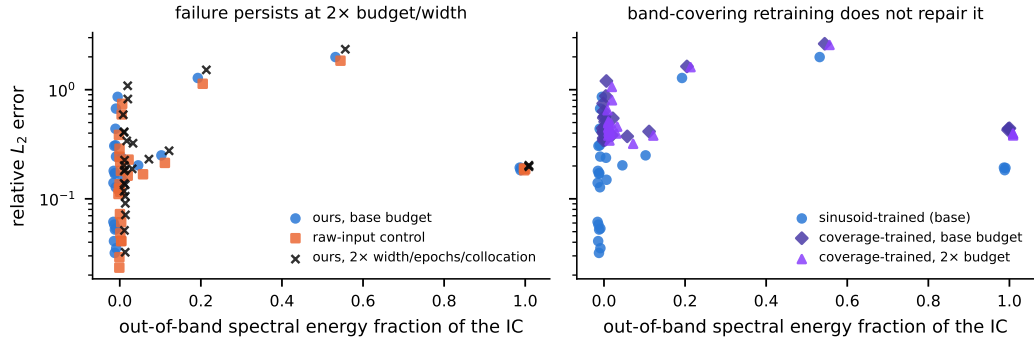


Figure 5: **The generalization boundary (reaction; held-out and probe ICs; log scale).** Each point is one IC's mean relative L_2 error against the fraction of its (non-mean) spectral energy outside the trained band (modes $\{1, 2\}$); series are offset horizontally for legibility. Left: the failure at high out-of-band content survives a simultaneous $2\times$ increase of encoder-path width, epochs, and collocation (crosses), which meanwhile improves in-band fitting—the failure is not under-training. Right: retraining on a distribution that genuinely covers modes ≤ 8 does not repair the probes at base budget (diamonds; it also degrades all in-band accuracy) nor at $2\times$ budget (triangles): with the solution network held fixed, coverage trades fit for span.

families is therefore interpolation in the learned coordinates, not extrapolation. The right panel makes this quantitative and predictive: a single scalar computed from the specification alone—distance of the IC's truncated spectrum to the trained set—rank-predicts held-out error almost perfectly (Spearman ≈ 0.95 for both arms) across three qualitatively different query types spanning two orders of magnitude of error. The same statistic explains the axis ordering of Table 1 (combinations $<$ parameters $<$ families: median distances grow in that order) and the width ladder within the Gaussian family (reaction, ours, across amplitudes: $\sigma=\pi/2$: 0.13 – 0.24 ; $3\pi/8$: 0.31 – 0.44 ; $\pi/4$: 0.67 – 0.86). Error is governed by distance-in-specification-space—which the next section pushes to its breaking point.

6.4 Where generalization stops: coverage sets the boundary, bandwidth caps it

The stress-probe set maps where specification conditioning stops working, and the answer is sharply organized by the same statistic that governs the smooth regime (Figure 5). Probes whose spectra leave the trained band fail in proportion to how much energy leaves it: narrow Gaussians fail outright ($\sigma=\pi/8$: 1.28; $\pi/16$: 1.99 mean relative L_2 error), while higher-harmonic sinusoids of modest amplitude degrade gracefully (0.18–0.19 at $\omega \in \{3, 4, 6\}$)—the model simply answers with the *in-band content* of the query (for $0.5 + 0.25 \sin 6x$ it predicts the logistic flow of the mean level 0.5, and its error is consistent with the ignored ripple’s share of the solution norm). Saturated sinusoids, mostly in-band, sit in between (0.20–0.25). Because the reaction references are closed form, no solver error contributes anywhere on this axis.

The failure is a boundary, not under-training. Doubling the parameter-encoder path width (64 $\rightarrow$ 128 branches, 150 $\rightarrow$ 300 fusion), training epochs (20k $\rightarrow$ 40k), and collocation density (1000 $\rightarrow$ 2000/specification) simultaneously—leaving the shared solution network untouched—*halves* seen-specification error (0.0099 $\rightarrow$ 0.0050) yet leaves every probe unchanged or slightly worse ($\sigma=\pi/16$: 2.35 vs. 1.99; ω -probes 0.20 vs. 0.19; Figure 5, left), and even degrades the in-family held-out sets somewhat (0.1453 vs. 0.1088)—extra conditioning capacity buys sharper fitting of the trained specifications, not broader generalization. The raw-input control shares the failure at matched budget, so it is a property of the conditioning problem, not of our encoder.

Band-covering retraining does not repair it—and the reason is diagnostic. The mechanism of §6.3 suggests an obvious mitigation: train on initial conditions that cover the band the probes occupy. We attempted this three times, each declared with predictions before running, and report the outcome as a finding. (1) A random Fourier-series distribution with modes ≤ 6 and $1/k^{1.5}$ amplitude decay leaves modes ≥ 3 at negligible amplitude—nominal coverage is not amplitude coverage—and changes nothing (probe mean 0.7731 vs. 0.6133). (2) A corrected distribution (modes ≤ 8 , flat decay) genuinely covers the band, but at the held-fixed budget it *trades fit for span*: the model can no longer fit even its own training distribution (seen 0.0756 vs. 0.0099 for the sinusoid-trained model), and every held-out and probe error worsens. (3) At $2\times$ budget and encoder width the coverage-trained model fits its own distribution better (seen 0.0431) yet *still* fails every probe (ω -probes 0.38–0.40—worse than the smooth-trained model’s in-band fallback of 0.19—and narrow Gaussians ≥ 1.5 ; Figure 5, right). Throughout, the band-limited $K=4$ encoder is statistically indistinguishable from $K=16$ (probe means 0.8445 vs. 0.8641 at $2\times$; 0.1247 vs. 0.1088 in the smooth regime), so the encoder’s representable band is never the binding constraint.

Together these results locate the boundary precisely: *not* in the specification encoder, but in the held-fixed solution pathway. Even when trained directly and exclusively on rough, band-covering specifications, the shared decoder fits them only to ≈ 0.04 –0.10 relative L_2 error—an order of magnitude above its accuracy on smooth families—so there is nothing accurate for the conditioning to generalize *to*: coverage cannot buy what capacity cannot fit. The out-of-distribution boundary of specification conditioning at this scale is a solution-capacity boundary sharpened by conditioning coverage: in the smooth regime it appears as the distance–error law of Figure 4; in the rough regime as a floor that neither doubled encoder capacity, band coverage, nor both together break. Enlarging the shared solution network is the natural next step, but changes the platform this study holds fixed.

6.5 One model across boundary conditions

Extending the encoder with the boundary-condition branch, a single reaction–diffusion model trains across periodic, Dirichlet, and zero-flux Neumann conditions—the boundary term of the loss expressing whichever BC each specification prescribes, with Dirichlet/Neumann references from the validated Crank–Nicolson solver (Appendix F). One property of the benchmark should be stated up front: the cosine-type ICs that make all three types jointly admissible are even about the domain endpoints, so by parity the periodic and zero-flux solutions *coincide* on this class (reference difference $\sim 10^{-6}$; the frozen baseline’s identical periodic/Neumann errors in Table 5 reflect this), whereas Dirichlet data genuinely changes the solution ($\approx 3.5\%$ relative L_2 error from periodic at $\nu=\rho=2$). The study therefore spans two analytically distinct boundary operators plus a third conditioned type that the model must learn to treat as equivalent—itsself a conditioning consistency test. The single model answers (Figure 6, full table in Appendix D): unseen ICs under periodic/Dirichlet/Neumann at 0.0640/0.0714/0.0643 (frozen coefficient-only: 0.2192/0.3113/0.2192); the 99 withheld (coefficient, IC, BC) pairings at 0.0792 (frozen: 0.2224)—BC-type composition generalizes essentially at

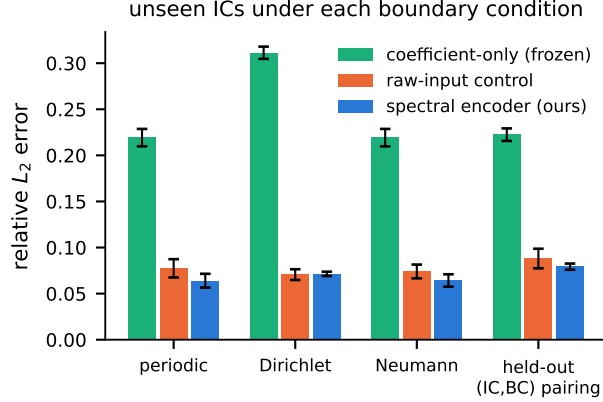


Figure 6: **Boundary-condition conditioning (reaction–diffusion, $(\nu, \rho) \in \{1, 2, 3\}^2$)**. Unseen ICs queried under each of the three conditioned BC types, and (coefficient, IC, BC) pairings never trained (each pair was trained under two of the three types; the withheld third is queried). Bars: mean $\pm$ std over 3 seeds.

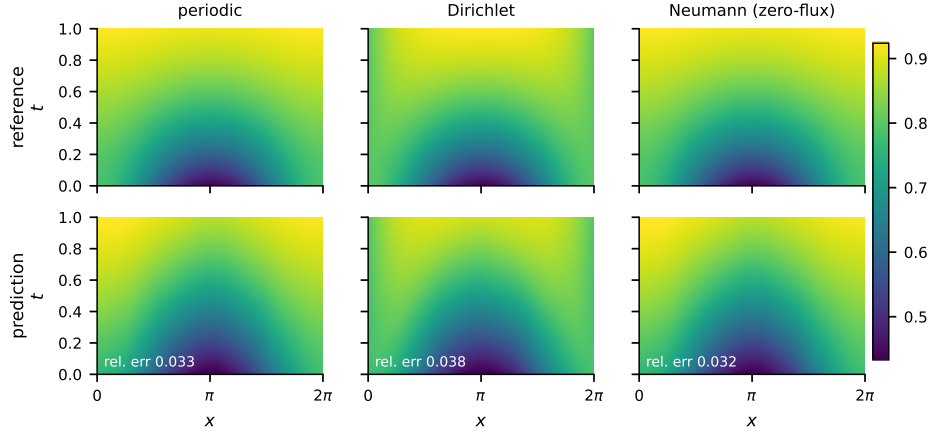


Figure 7: **One model, one unseen IC, three boundary conditions** (reaction–diffusion, $\nu=\rho=2$, seed 0): the same query profile under periodic, constant-Dirichlet (values conditioned from the IC endpoints), and zero-flux Neumann conditions, against the spectral (periodic) and Crank–Nicolson (non-periodic) references.

seen-specification accuracy (0.0783); and never-trained IC families under Dirichlet data at 0.3248 (frozen: 0.8429). Dirichlet queries at unseen ICs entail unseen boundary *values* (the conditioned g varies continuously with the IC), so value, type, and profile generalize jointly. The raw-input control again ties (0.0706, 0.0881); the anchor cost repeats (specialist 0.0575 vs. 0.0707). Figure 7 shows one unseen IC solved under all three BC types by one model, against the three references.

6.6 One model across operators, and the limits of composition

One model trained across convection, reaction, and reaction–diffusion specifications—the coefficient vector itself selecting the operator, with no family token—supports all three at once (Table 2): seen specifications at 0.0439, unseen ICs within each family at 0.1710, unseen coefficient–IC combinations at 0.1006, and unseen IC families at 0.4583, mirroring the single-family studies. The most speculative queries are *operator compositions* never trained in any form: convection–diffusion and full CDR coefficient settings, whose reference solutions we refinement-check before use (Appendix F). Transfer is partial and honest to report as such: 0.2338 on convection–diffusion and 0.2921 on CDR (per-composition means 0.20–0.33, with large seed variance up to ± 0.15)—clearly structured (each

Table 2: **A single model across three operator families** (convection $\beta \leq 5$, reaction $\rho \leq 5$, reaction-diffusion $\nu, \rho \leq 2$; the coefficient vector itself selects the operator). relative L_2 error, mean $\pm$ std over 3 seeds. The last two rows are operator compositions never trained in any form.

Evaluation	Raw-input control	Spectral encoder (ours)
seen specifications (3 families)	0.0426 $\pm$ 0.0044	0.0439 $\pm$ 0.0027
unseen ICs (per family)	0.1677 $\pm$ 0.0104	0.1710 $\pm$ 0.0144
unseen IC family	0.4633 $\pm$ 0.0485	0.4583 $\pm$ 0.0868
unseen (coeff, IC) combinations	0.0858 $\pm$ 0.0088	0.1006 $\pm$ 0.0001
convection-diffusion (family never trained)	0.2115 $\pm$ 0.0224	0.2338 $\pm$ 0.0026
CDR (family never trained)	0.2575 $\pm$ 0.0355	0.2921 $\pm$ 0.0911

conditioned prediction tracks the right IC and qualitative dynamics; errors sit between the within-family unseen-IC level of 0.1710 and the unseen-family level of 0.4583) but far from within-family accuracy. Compositions queried at ICs that are *also* unseen—specifications novel in operator, coefficients, and IC simultaneously—degrade only mildly further (CDR: 0.2675 vs. 0.3166 at trained ICs). The raw-input control is again statistically indistinguishable (0.2115, 0.2575). We conclude that coefficient-encoded operator identity composes to nearby unseen operators at usable but degraded accuracy; making such composition reliable is an open problem this platform can now measure.

7 Limitations

Our study inherits the scale of its platform: 1D CDR physics, a 77–86k-parameter surrogate, and full-batch first-order training. This is a deliberate choice—it makes matched-budget attribution and mechanism analysis affordable and exact references available—but the spectral representation as used here presumes a periodic-friendly domain and a scalar field; richer geometries would need a different (e.g., learned or graph-based) specification basis, and we do not test whether the coverage account of generalization survives that substitution. The boundary-condition axis covers three conditioned types with constant Dirichlet data—and, on the symmetric IC class that admits all three types, only two analytically distinct boundary operators (§6.5); time-varying or asymmetric boundary data would require extending both the representation and the authored reference solver. Operator breadth is demonstrated within one scalar PDE family whose members share the CDR form; the weak transfer we observe to held-out operator compositions (§6.6) suggests—but on this evidence only suggests—that specification conditioning alone does not deliver new-operator generalization. Finally, the coefficient-only baseline is evaluated frozen on held-out specifications because it is not queryable there; per-specification retraining would eventually beat any amortized model at proportionally higher cost, a trade-off we quantify only through the single-model setting.

8 Conclusion

Parameterized physics-informed surrogates need not stop at the coefficient vector. Conditioning the parameter encoder on the initial condition and a typed boundary embedding turns a coefficient-indexed PINN into a specification-indexed one: a single model, trained once under the original objective and budget, that answers queries across coefficients, initial conditions (including functional families it never saw), boundary-condition types, and several operators of a PDE family, at a fraction of the error of the frozen coefficient-only surrogate that is the actual alternative at equal cost. The gains are causal—corrupting the conditioning removes them—and legible: the encoder’s latent space organizes initial conditions by their low-order spectrum, and a scalar computed from the specification alone predicts where the model will fail. The study’s most instructive result is the one we did not plan: at matched capacity, an unstructured raw-sample conditioning interface generalizes as well as our spectral, factorized one wherever the two carry the same information, and out-of-band queries fail regardless of encoder capacity or truncation, and band-covering retraining cannot repair them because the held-fixed solution network cannot accurately fit rough specifications even when supervised on them directly—the boundary of specification conditioning is set by what the shared solution pathway can represent, and sharpened by what the training distribution covers. For amortized physics-informed solvers, we read this as a shift in the design question: from *how to architect* the conditioning pathway

to *what the specification representation covers*—its information, its bandwidth, and the region of specification space the training distribution actually spans.

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A Artifacts and reproducibility

All experiments in this paper are generated by the accompanying code (proposal/code/): the specification/split definitions (with the exact IC lists and seeds fixed in a dated plan file and machine-readable split dumps written before the comparison runs, and amendments dated before the runs they introduce), the training and evaluation harness, the authored non-periodic solver and its validation suite, and the analysis scripts that produce every table, figure, and in-text number directly from the per-run logs (proposal/logs/runs/, one directory per run with its configuration, specification lists, training curve, model weights, and per-specification metrics). The reference P²INN implementation [Cho et al., 2024] is reused verbatim for everything held fixed and is cross-validated against the authors’ released checkpoint (Appendix E).

B Non-regression at the anchor specification, per coefficient

Per-coefficient relative L_2 error at the anchor IC (the coefficient-only model’s trained setting), 3-seed means. “Limit” is the pre-registered tolerance (baseline + $\max(10\%, 0.005)$); cells above it are marked. Covering the full specification space at the specialist’s exact budget costs accuracy at the specialist’s own specification in some cells; we report every cell.

Table 3: Anchor-specification relative L_2 error per coefficient: reaction (ρ ; top block), reaction–diffusion ((ν, ρ) ; second block), convection (β ; third block), and the boundary-condition-study model at its periodic anchor ((ν, ρ) ; bottom block).

Coefficient	Coefficient-only	Raw-input	Spectral (ours)	Limit	Within?
1	0.0048	0.0058	0.0121	0.0098	×
2	0.0040	0.0050	0.0098	0.0090	×
3	0.0032	0.0038	0.0082	0.0082	✓
4	0.0027	0.0034	0.0070	0.0077	✓
5	0.0025	0.0028	0.0057	0.0075	✓
(1,1)	0.0600	0.0937	0.0958	0.0660	×
(1,2)	0.0399	0.0667	0.0640	0.0449	×
(1,3)	0.0317	0.0474	0.0450	0.0367	×
(2,1)	0.0934	0.1452	0.1462	0.1027	×
(2,2)	0.0648	0.1039	0.0997	0.0713	×
(2,3)	0.0543	0.0773	0.0709	0.0597	×
(3,1)	0.1132	0.1704	0.1706	0.1245	×
(3,2)	0.0821	0.1262	0.1196	0.0903	×
(3,3)	0.0725	0.0960	0.0856	0.0798	×
1	0.0084	0.0158	0.0131	0.0134	✓
2	0.0071	0.0136	0.0158	0.0121	×
3	0.0073	0.0162	0.0180	0.0123	×
4	0.0070	0.0197	0.0168	0.0120	×
5	0.0076	0.0205	0.0233	0.0126	×
(1,1)	0.0521	0.0587	0.0469	0.0573	✓
(1,2)	0.0312	0.0502	0.0395	0.0362	×
(1,3)	0.0228	0.0390	0.0331	0.0278	×
(2,1)	0.0868	0.1031	0.0874	0.0954	✓
(2,2)	0.0534	0.0882	0.0730	0.0587	×
(2,3)	0.0386	0.0731	0.0622	0.0436	×
(3,1)	0.1114	0.1316	0.1106	0.1225	✓
(3,2)	0.0698	0.1196	0.0976	0.0768	×
(3,3)	0.0518	0.1000	0.0857	0.0570	×

C Absolute errors, convection curves, raw-control ablations

The raw-input control’s conditioning ablations mirror Figure 3: on reaction, shuffled-IC evaluation degrades its unseen-IC-parameter error from 0.0909 to 0.2700 and constant-IC to 0.2041; on unseen families from 0.3444 to 0.4382 (shuffled) and 0.3999 (constant). Both arms therefore genuinely condition on the specification.

D Boundary-condition study, full table

Table 4: Absolute (L_1 -mean) error companion to Table 1, same protocol.

Method	Family	Seen spec.	Unseen IC params	Unseen IC family	Unseen combos	Budget
P ² INN (coefficient-only, frozen)	reaction	0.0023 $\pm$ 0.0015	0.1341 $\pm$ 0.0005	0.2661 $\pm$ 0.0010	0.1256 $\pm$ 0.0004	reference
raw-input conditioning (control)	reaction	0.0029 $\pm$ 0.0004	0.0483 $\pm$ 0.0040	0.1422 $\pm$ 0.0198	0.0131 $\pm$ 0.0050	matched
spectral specification encoder (ours)	reaction	0.0060 $\pm$ 0.0028	0.0576 $\pm$ 0.0032	0.1587 $\pm$ 0.0080	0.0303 $\pm$ 0.0071	matched
P ² INN (coefficient-only, frozen)	reaction–diffusion	0.0389 $\pm$ 0.0023	0.1209 $\pm$ 0.0020	0.2093 $\pm$ 0.0023	0.1194 $\pm$ 0.0024	reference
raw-input conditioning (control)	reaction–diffusion	0.0580 $\pm$ 0.0012	0.0550 $\pm$ 0.0039	0.0914 $\pm$ 0.0157	0.0562 $\pm$ 0.0012	matched
spectral specification encoder (ours)	reaction–diffusion	0.0557 $\pm$ 0.0014	0.0512 $\pm$ 0.0042	0.1061 $\pm$ 0.0061	0.0553 $\pm$ 0.0009	matched
P ² INN (coefficient-only, frozen)	convection	0.0078 $\pm$ 0.0031	0.8383 $\pm$ 0.0017	0.7924 $\pm$ 0.0022	0.8787 $\pm$ 0.0016	reference
raw-input conditioning (control)	convection	0.0215 $\pm$ 0.0026	0.3622 $\pm$ 0.0117	0.3693 $\pm$ 0.0081	0.2226 $\pm$ 0.0351	matched
spectral specification encoder (ours)	convection	0.0233 $\pm$ 0.0055	0.3615 $\pm$ 0.0588	0.3740 $\pm$ 0.0366	0.2678 $\pm$ 0.0308	matched

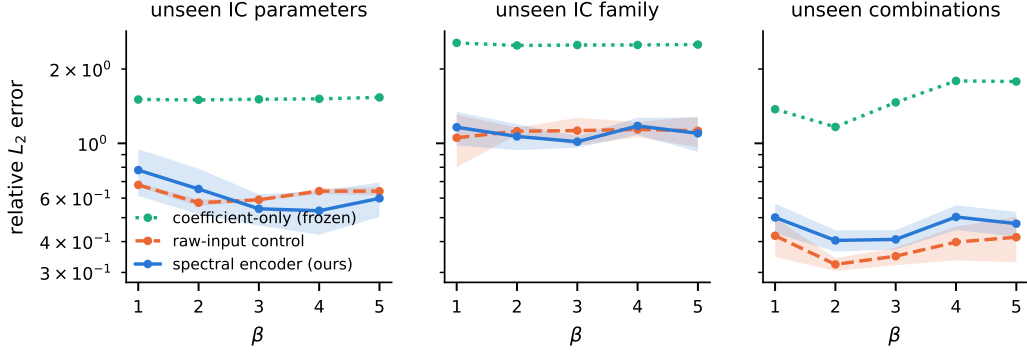


Figure 8: Error across the specification space for the convection family (companion to Figure 1). Transport makes conditioned generalization much harder in absolute terms for both conditioned arms (note the scale), while the frozen coefficient-only model is off the top of the useful range on every held-out axis.

E Reproduction of the reference model and pipeline cross-validation

Solver parity. Our solvers generalize the reference constructions only in accepting a callable u_0 ; on the reference IC zoo they agree with the original implementations to machine precision (max absolute difference 0.0 across the reaction, convection–diffusion, reaction–diffusion, and CDR solvers on six zoo ICs and representative coefficients).

Checkpoint cross-check. Loading the publicly released P²INN all-families checkpoint into our model class and evaluating it with *our* data and metric pipeline reproduces the published per-family errors to four decimals (relative L_2 error): convection 0.0316 (published 0.0316), diffusion 0.1694 (0.1694), reaction 0.0229 (0.0229), reaction–diffusion 0.1015 (0.1015), CDR 0.0508 (0.0508). Our data generation, model definition, and evaluation are therefore exactly the reference’s.

Retrained reference baselines. Training the coefficient-only model from scratch in the published Gaussian-IC setting yields, over 3 seeds, relative L_2 error 0.0020 ± 0.0003 on reaction $\rho \in [1, 5]$ (published 0.0027 ± 0.0030) and 0.0062 ± 0.0020 on convection $\beta \in [1, 5]$ (published 0.0079 ± 0.0008)—consistent with the paper. Our anchor-IC baselines (Table 1) are trained identically with the anchor initial profiles of §5.

F Authored non-periodic reference solver and its validation

For Dirichlet and Neumann reaction–diffusion references we use the same Lie splitting as the reference periodic solver—an exact logistic reaction step followed by a diffusion step per $\Delta t = 10^{-2}$ —replacing the spectral diffusion step with Crank–Nicolson finite differences on a 513-node grid containing the reference evaluation points ($4\times$ the evaluation resolution in space), with the BC imposed in the scheme (pinned values for Dirichlet; second-order mirrored ghost nodes for zero-flux Neumann). Validation (full records in `logs/solver_validation/`):

Table 5: Reaction–diffusion boundary-condition study, relative L_2 error (mean $\pm$ std, 3 seeds). The coefficient-only model is trained at the anchor IC under periodic conditions and evaluated frozen elsewhere. Unseen-IC rows: 6 fresh cosine-type ICs per BC type (54 specifications each); the (coeff, IC, BC) row: the 99 withheld type pairings; unseen-family rows: Gaussians/bumps under periodic and Dirichlet data (the bump/Gaussian endpoint slopes are not zero-flux-compatible, so Neumann is excluded there by construction). By parity of the cosine-type ICs, the periodic and zero-flux reference solutions coincide on this class, which is why the frozen baseline’s periodic and Neumann rows match; Dirichlet is the analytically distinct operator.

Evaluation	Coefficient-only (frozen)	Raw-input	Spectral (ours)
seen (anchor, periodic)	0.0575 $\pm$ 0.0077	0.0848 $\pm$ 0.0194	0.0707 $\pm$ 0.0067
seen specifications	0.2039 $\pm$ 0.0067	0.0889 $\pm$ 0.0120	0.0783 $\pm$ 0.0038
held-out (coeff, IC, BC) pairings	0.2224 $\pm$ 0.0068	0.0881 $\pm$ 0.0106	0.0792 $\pm$ 0.0033
unseen ICs, periodic	0.2192 $\pm$ 0.0094	0.0775 $\pm$ 0.0099	0.0640 $\pm$ 0.0075
unseen ICs, Dirichlet	0.3113 $\pm$ 0.0067	0.0706 $\pm$ 0.0058	0.0714 $\pm$ 0.0023
unseen ICs, Neumann	0.2192 $\pm$ 0.0094	0.0741 $\pm$ 0.0074	0.0643 $\pm$ 0.0067
unseen IC family, periodic	0.4853 $\pm$ 0.0037	0.2101 $\pm$ 0.0044	0.2396 $\pm$ 0.0687
unseen IC family, Dirichlet	0.8429 $\pm$ 0.0090	0.2364 $\pm$ 0.0374	0.3248 $\pm$ 0.0426

- *Periodic overlap*: replacing only the diffusion step of the reference operator-split solver with the CN machinery (periodic circulant form, $4\times$ substeps) reproduces the reference spectral solution to $\leq 3.1 \times 10^{-4}$ maximum relative deviation at the hardest setting ($\nu=\rho=3$).
- *Self-convergence*: refining space and time together ($513 \rightarrow 1025 \rightarrow 2049$ nodes, $\Delta t \rightarrow \Delta t/2 \rightarrow \Delta t/4$) changes the Dirichlet/Neumann solutions by at most 4.3×10^{-3} (coarse vs. refined) and 1.0×10^{-3} (refined vs. doubly refined), decreasing at first order as expected for Lie splitting—an order of magnitude below the margins the study interprets.
- *Operator-split reference convergence*: at the sharpest evaluated periodic reaction–diffusion and CDR specifications, doubling the spatial resolution and halving the splitting step changes the reference test values by $< 10^{-3}$ relative L_2 error-equivalent, below our decision floor of 0.015.

G Spectral truncation selection

K was selected on the validation points (the reference validation indices) of the *trained* specifications of the reaction study, structured arm, seed 0, full budget—no held-out specification was involved. Mean relative L_2 error: $K=4$: 0.0134; $K=8$: 0.0106; $K=16$: 0.0071. The pre-registered rule (smallest K within 10% of the best) selects $K=16$. Because the trained sinusoids contain no energy above mode 2, the differences reflect optimization and initialization effects of the wider input layer rather than information content; the band-limited $K=4$ variant is retained as a diagnostic in §6.4.